Morteza Poudat

Senior Data Engineer / Python Developer

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PROFILE

Senior Data Engineer and Python Developer with 9+ years of experience building data pipelines, ETL processes and backend applications for engineering and materials data. Specialised in turning unstructured technical sources (PDF handbooks, scanned tables, Excel, XML, JSON, SQL) into validated, well-modelled databases, including PDF parsing, table detection and OCR. Background in computational mechanics and civil engineering (MSc), combined with hands-on customer consulting and implementation project management. Also builds full-stack web applications (React/TypeScript, PostgreSQL). Daily user of AI coding agents and LLM-assisted development workflows; writes clean, reproducible, documented production code.

CORE SKILLS

- **Languages & Data:** Python (Pandas), SQL, Matlab, C++
- **Databases & Modelling:** MongoDB, PostgreSQL, data modelling, schema and naming-rule design, data migration
- **ETL & Extraction:** ETL pipeline design, PDF parsing (PyMuPDF), table-grid detection (OpenCV), OCR (Tesseract), web scraping, data validation
- **Backend & DevOps:** Django, FastAPI, REST/API integration, Docker, Linux, supervisord, Git/GitLab (merge requests, code review), CI/CD
- **Frontend & Testing:** React, TypeScript, Next.js, Vite, Vitest, Python tests, flake8, oxlint
- **Analytics & AI:** Apache Superset; AI-agent / LLM coding workflows (Cursor)
- **Engineering:** FEM / computational mechanics (Abaqus incl. UMAT, ANSYS APDL), materials data (steel, copper, aluminium alloys)

PROFESSIONAL EXPERIENCE

Senior Data Engineer & Senior Application Consultant Jun 2017 – Present

Engineering materials-data software company, Germany

- **Developed full-stack web applications and plugins for the company's materials data platform:** Python backend, PostgreSQL (Docker, migrations, SQL-side reporting), React/TypeScript/Vite frontend, automated tests (Vitest, Python), GitLab CI/CD, role-based permissions, English/German i18n; AI-assisted development with Cursor.
- **Materials data modelling:** Designed a unified materials catalogue data model in MongoDB for steel, copper and aluminium, including classification and source trees, property naming rules and chemical composition rules. Rebuilt the aluminium database, including its property definitions.
- **PDF extraction & ETL:** Built extraction and ETL pipelines for technical handbooks and design guidelines using PyMuPDF, OpenCV table-grid detection and two-engine per-cell OCR (Tesseract). Parsed a fatigue-design guideline into linked records (tests, specimens and load spectra linked to materials); benchmarked the pipeline against MinerU.
- **ETL standardisation:** Defined a standard 7-step ETL design (read, extract/OCR, map to generic properties, validate, package, load, review) and packaged reproducible pipelines in Git/GitLab with READMEs.
- **Further data imports:** Time-temperature-transformation, metallography, life-cycle and copper datasets.
- **Customer data migrations:** Migrated customer data into the company's materials data platform with Python importers and master-material matching tables, delivered via GitLab merge requests.
- **Senior Application Consultant:** Worked closely with customers in workshops and ongoing discussions to analyse requirements and design solutions; developed custom plugins for the company's materials data platform tailored to customer requirements; provided customer support from bug report to delivered feature.
- **Project management:** Implementation project management, including weekly customer meetings and cost proposals for a multi-phase data platform.
- **Knowledge sharing:** Conference talk on copper alloys data (2026) and a submitted abstract on materials data.

Civil Engineering Intern 2009 – 2010
Iran

SELECTED FREELANCE WORK

- **Bilingual (DE/EN) portfolio website for a Düsseldorf artist** — HTML, CSS, JavaScript (live website)
- **Online booking system for a barbershop (client prototype)** — Next.js 16, TypeScript, PostgreSQL; bilingual (DE/EN), mobile-first booking, confirmation e-mails with calendar file, admin with two-factor login

EDUCATION

M.Sc. Computational Mechanics 2016 – 2021
University of Duisburg-Essen, Germany · Grade 1.9

- Thesis: Knowledge base for the selection and application of aluminium alloys.

Materials Science & Applied Mechanics (semesters) 2015 – 2016
University of Duisburg-Essen, Germany

M.Sc. Civil Engineering
Shahid Chamran University of Ahvaz, Iran

- Finite element simulation in Abaqus, including user material subroutines (UMAT).

B.Sc. Civil Engineering
Shahid Chamran University of Ahvaz, Iran

CERTIFICATES

- Real Python – Pandas (2021–2022)
- Sololearn – Python for Data Science; Intermediate Python (2021)
- MongoDB – MongoDB Basics (2018)
- MITx 6.00.1x – Introduction to Computer Science and Programming Using Python (2017)

LANGUAGES

Persian – native | German – full professional proficiency | English – fluent